Fundamentals of Machine Learning

Understanding Test, Lectures 1 to 4

Chapter by chapter: true-or-false statements and open questions covering every concept, with the answer key at the end.

How to use this test

Four chapters, one per lecture. Each has Section A (true or false, each needing a one-line reason) and Section B (open questions). Do a chapter closed-book in about 40 minutes, then mark yourself against the answer key, which starts on page 9. A statement is *false* if any part of it is false; the key says which part. Pointers [one-read §x] refer to the one-read document.

Contents

1 Chapter 1: What does it mean to learn? **1**

2 Chapter 2: From representations to decisions **3**

3 Chapter 3: From fitting to generalisation **4**

4 Chapter 4: Discriminative and generative classification **6**

Answer key **9**

Answer key, Chapter 1 **9**

Answer key, Chapter 2 **11**

Answer key, Chapter 3 **14**

Answer key, Chapter 4 **18**

1 Chapter 1: What does it mean to learn?

Section A: true or false

Mark each statement true or false and give a one-line reason. A verdict without a reason scores zero.

A1. T ☐ F ☐ Learning requires two things: experience, and a capacity to change in response to it.

A2. T ☐ F ☐ Having an ability implies being able to write down the procedure that produces it.

A3. T ☐ F ☐ Expert systems failed because computers were too slow to run their rules.

A4. T ☐ F ☐ Encoding human knowledge as rules is obsolete now that machines can learn.

A5. T ☐ F ☐ In supervised learning the experience consists of inputs paired with desired outputs.

A6. T☐ F☐ Self-supervised learning requires a human to label a small subset of the data.

A7. T☐ F☐ In reinforcement learning the learner's actions change the experience it receives afterwards.

A8. T☐ F☐ Whether a problem is regression or classification is decided by the type of the inputs.

A9. T☐ F☐ A prediction, a decision and an action are three different things.

A10. T☐ F☐ Representation learning removes the need for human choices in building a system.

A11. T☐ F☐ An autoregressive language model is trained by supervised learning on human-written answers.

A12. T☐ F☐ Of the four ingredients of a learning system, the evaluation criterion decides which changes to the model count as improvements.

A13. T☐ F☐ A system that classifies all of its training examples correctly has demonstrated generalisation.

A14. T☐ F☐ Cross-entropy measures how good predicted probabilities are.

Section B: open questions

Answer in two to five sentences, or a short derivation where asked. Precision matters more than length.

B1. State the two ways of giving a machine an ability, and give one task where each is the right choice.

B2. Give the three forms of experience, the learning setting each leads to, and what provides the learning signal in each.

B3. Labels can come from three kinds of sources. Name them and give one danger of each.

B4. Describe the supervised training cycle in four steps.

B5. A hospital model outputs a probability of disease. Identify the prediction, a decision, and an action, and explain why the probability alone does not determine the action.

B6. What is the difference between feature engineering and representation learning? What did deep learning change?

B7. Explain the "wolves and snow" failure using the vocabulary of the chapter, and say what experiment reveals it.

B8. Explain in one paragraph how generation emerges from a model that was only ever trained to predict.

2 Chapter 2: From representations to decisions

Section A: true or false

Mark each statement true or false and give a one-line reason. A verdict without a reason scores zero.

A1. T☐ F☐ A categorical feature with five values should be encoded as the integers 1 to 5 before being used with Euclidean distance.

A2. T☐ F☐ If the conditional distribution $P(Y \mid X = x)$ were known, the best decision would still depend on the loss.

A3. T☐ F☐ Under squared loss the optimal prediction is the conditional median.

A4. T☐ F☐ With a cost c_u per unit of under-prediction and c_o per unit of over-prediction, the optimal prediction is the quantile of $Y \mid x$ at level $c_u/(c_u + c_o)$.

A5. T☐ F☐ For a binary classifier with costs C_{FP} and C_{FN}, the optimal rule declares positive when $p(1 \mid x) > C_{FN}/(C_{FP} + C_{FN})$.

A6. T☐ F☐ Adding a reject (review) option can never change the decision on a case that was already clearly positive.

A7. T☐ F☐ The i.i.d. assumption says that each example is drawn from the same distribution as the deployment cases, independently of the others.

A8. T☐ F☐ In k-NN, a larger k gives the model more capacity.

A9. T☐ F☐ Scaling the features changes the predictions of k-NN but not those of a decision based on the true posterior.

A10. T☐ F☐ In high dimension, almost all of a query's n training points are at nearly the same distance from it, so "nearest" loses meaning.

A11. T☐ F☐ A hypothesis space is the set of functions a parameterised model can represent.

A12. T☐ F☐ The number of neighbours k is a parameter of the k-NN model.

A13. T☐ F☐ The softmax output of a model with C classes lies in the probability simplex Δ^{C-1}.

A14. T☐ F☐ Empirical risk minimisation chooses the parameters that minimise the population risk.

A15. T☐ F☐ Cross-entropy is used as a surrogate because the 0–1 loss has zero gradient almost everywhere.

A16. T☐ F☐ The validation set is used to fit the parameters and the test set to choose the hyperparameters.

Section B: open questions

Answer in two to five sentences, or a short derivation where asked. Precision matters more than length.

B1. Derive the Bayes decision rule for the 0–1 loss from the conditional risk.

B2. With $C_{FP} = 1$ and $C_{FN} = 4$, derive the posterior threshold and explain why it is below 0.5.

B3. Explain why two training examples with identical x and different y put a floor on the error of every classifier, and name that floor.

B4. Describe how k-NN estimates $P(Y = c \mid x)$ and what the estimate's bias and variance depend on.

B5. Explain the curse of dimensionality in terms of a hypercube with cells of side 0.1, and what it implies for local methods.

B6. Contrast the two strategies of the chapter, local estimation and parameterised models, in terms of what each assumes and what each needs.

B7. Write the binary cross-entropy loss for a predicted probability $\hat{p}$ and label y, and explain what minimising its average over a training set does.

B8. Why does a parameterised model's prediction at an input far from all training points still exist, and why is it suspect?

B9. Give the role of each of the three subsets and state the one rule that must never be broken.

B10. What is a hyperparameter, how does it differ from a parameter, and give three examples from this chapter.

3 Chapter 3: From fitting to generalisation

Section A: true or false

Mark each statement true or false and give a one-line reason. A verdict without a reason scores zero.

A1. T☐ F☐ The generalisation gap is the difference between the population risk and the empirical risk of the same fitted model.

A2. T☐ F☐ On average, the training risk of the ERM solution is below even the best population risk achievable in the family.

A3. T☐ F☐ A test set gives an unbiased estimate of a model's risk provided the model was fitted without it, even if the same test set was then used to choose between ten such models.

A4. T☐ F☐ For nested families $\mathcal{H}_1 \subseteq \mathcal{H}_2$, the minimum training error over $\mathcal{H}_2$ cannot exceed that over $\mathcal{H}_1$.

A5. T☐ F☐ Capacity is the number of parameters of the model.

A6. T☐ F☐ Underfitting is recognised by low training error and high validation error.

A7. T☐ F☐ In the bias–variance decomposition, the noise term can be reduced by collecting more data.

A8. T☐ F☐ Variance measures how much the fitted prediction at a fixed x changes when the training set is redrawn.

A9. T☐ F☐ Approximation error is reduced by enlarging the hypothesis space; estimation error is reduced by more data.

A10. T☐ F☐ An L_2 penalty on the parameters reduces capacity by removing parameters from the model.

A11. T☐ F☐ An L_1 penalty tends to set some coefficients exactly to zero.

A12. T☐ F☐ Early stopping controls capacity even though the number of parameters never changes.

A13. T☐ F☐ Data augmentation is a way of adding information about which variations should not change the output.

A14. T☐ F☐ Double descent means the validation error always decreases if capacity is increased far enough.

A15. T☐ F☐ Forward selection and backward elimination are guaranteed to find the best feature subset.

A16. T☐ F☐ Selecting the configuration with the lowest cross-validation error and reporting that error as the final performance is sound.

A17. T☐ F☐ A stratified split protects against near-duplicate flows from one attack episode appearing on both sides.

A18. T☐ F☐ A temporal split is appropriate when the deployment cases come from the future.

A19. T☐ F☐ A classifier with accuracy 0.99 on a dataset with 1% positives must have caught most positives.

A20. T☐ F☐ Recall and false-positive rate do not depend on the class prevalence, but precision does.

A21. T☐ F☐ Computing the standardisation statistics on the full dataset before splitting is leakage even though no labels are used.

Section B: open questions

Answer in two to five sentences, or a short derivation where asked. Precision matters more than length.

B1. Explain why training data are “used twice” and why this makes the training risk optimistic.

B2. List the four “doors” through which inductive bias enters a learning system, and explain why some inductive bias is unavoidable.

B3. Write the bias–variance–noise decomposition of the expected squared error at a fixed x and define each term.

B4. A model shows training error 0.02 and validation error 0.25. Diagnose, list three remedies, and say what you would expect from collecting ten times more data.

B5. Explain geometrically why L_1 regularisation produces sparse solutions and L_2 does not.

B6. What is patience in early stopping, and what goes wrong if it is too short or too long?

B7. Two configurations have the same population error R and validation errors $R + \varepsilon_i$ with independent $\varepsilon_i \sim N(0, \sigma^2)$. Show that the expected validation error of the selected one is $R - \sigma/\sqrt{\pi}$ and interpret.

B8. For each of the following deployment goals, name the right split: (i) new flows from the same process; (ii) next month's traffic; (iii) new attack episodes; (iv) a rare class whose counts must be stable across folds.

B9. Define precision, recall, specificity and F_1, and give one question each answers.

B10. List five forms of leakage and give the rule that prevents the preprocessing kind.

B11. Describe the learning-curve patterns that indicate (a) underfitting, (b) high variance, (c) a data-limited problem, and the response to each.

4 Chapter 4: Discriminative and generative classification

Section A: true or false

Mark each statement true or false and give a one-line reason. A verdict without a reason scores zero.

A1. T☐ F☐ Once the posterior $p(y \mid x)$ is known, the decision is easy; the hard part of classification is estimating the posterior.

A2. T☐ F☐ A discriminative model estimates $p(x \mid y)$ and $p(y)$ and combines them with Bayes' rule.

A3. T☐ F☐ In logistic regression the score is $w^\top x + b$ and the posterior estimate is the sigmoid of the score.

A4. T☐ F☐ Minimising cross-entropy in logistic regression is the same as maximising the conditional likelihood of the labels.

A5. T☐ F☐ Changing the cost threshold rotates the decision boundary of a logistic model.

A6. T☐ F☐ A generative model can in principle generate new inputs; a discriminative model cannot.

A7. T☐ F☐ Maximum likelihood for a Gaussian class model sets the class mean to the sample mean and the covariance to the sample covariance of that class.

A8. T☐ F☐ LDA assumes each class has its own covariance matrix.

A9. T☐ F☐ The LDA decision boundary is linear because the quadratic terms in x cancel between classes.

A10. T☐ F☐ In d dimensions, QDA with two classes estimates about $d(d+1)$ covariance parameters, twice as many as LDA.

A11. T☐ F☐ Covariance shrinkage blends the estimated covariance with a simpler matrix to reduce estimation variance.

A12. T☐ F☐ Naive Bayes assumes that the features are independent.

A13. T☐ F☐ Without smoothing, one feature value never seen in a class makes that class's posterior exactly zero.

A14. T☐ F☐ Correlated features make Naive Bayes under-confident.

A15. T☐ F☐ A generative model can handle a missing feature by marginalising it out; a discriminative model needs a separate treatment.

A16. T☐ F☐ A flexible discriminative model can eventually reach lower error than a generative one because it spends no capacity on aspects of x irrelevant to y.

A17. T☐ F☐ If the training and deployment distributions are identical, an imbalanced prior is an error that must be corrected by resampling.

A18. T☐ F☐ A classifier is calibrated if, among cases given probability u, about a fraction u are positive.

A19. T☐ F☐ Under class-prior shift the class-conditional distributions change and the priors stay the same.

A20. T☐ F☐ A high maximum posterior $\max_c \hat{p}(c \mid x)$ shows that x resembles the training data.

A21. T☐ F☐ A per-flow classifier cannot detect an attack whose individual flows all look ordinary, but a classifier on window features can.

A22. T☐ F☐ An autoregressive language model is trained generatively, and a sequence-classification head on top of it is trained discriminatively.

Section B: open questions

Answer in two to five sentences, or a short derivation where asked. Precision matters more than length.

B1. Write the two routes to the posterior as formulas and name what each must estimate.

B2. Explain why the sigmoid is the right link for a binary discriminative model, and the softmax for a multiclass one.

B3. Starting from the Gaussian log-posterior, explain which terms are common to all classes in LDA and write the resulting score $s_c(x)$.

B4. How does the class prior move the LDA boundary, and why does it not change the class models?

B5. State the Naive Bayes assumption as a formula, explain why it is attractive with little data, and give one way it fails for network-flow features.

B6. Give the discriminative/generative comparison on five criteria: modelled quantity, input assumptions, missing features, sampling inputs, raw high-dimensional input.

B7. Explain the difference between correcting for class imbalance in training and correcting for prior shift at deployment.

B8. Derive the prior-shift correction for a binary discriminative model from Bayes' rule, and compute it for $\pi_t = 0.2$, $\pi_d = 0.05$, $r_t(x) = 0.6$.

B9. Describe how a generative model can reject an unfamiliar input, and give one limitation of the method.

B10. A detector computes, in a 60-second window ending at each flow completion, the number of flows, distinct destination ports, failure fraction and score sum. State what information is legitimately available at each decision time and give two ways such a system leaks the future.

B11. Give the three questions the chapter uses for choosing between a generative and a discriminative model.

Answer key

Answer key, Chapter 1: What does it mean to learn?

True or false

A1. *Learning requires two things: experience, and a capacity to change in response to it.*
True. That is the course's definition: capacity to learn + experience → acquired abilities [one-read §1.1].

A2. *Having an ability implies being able to write down the procedure that produces it.*
False. People recognise a handwritten "3" in every style yet cannot state the rule; this gap is why rule-writing fails for perception and why learning from experience is needed [one-read §1.1].

A3. *Expert systems failed because computers were too slow to run their rules.*
False. They failed at the *knowledge acquisition bottleneck*: experts cannot articulate complete rules, and every rule meets exceptions. Speed was not the problem [one-read §1.1].

A4. *Encoding human knowledge as rules is obsolete now that machines can learn.*
False. It remains the right approach whenever the procedure is known and precise (arithmetic, protocol parsing). Learning is for abilities that cannot be specified [one-read §1.1].

A5. *In supervised learning the experience consists of inputs paired with desired outputs.*
True. Examples (x, y); the label may come from measurement, annotation, or the future [one-read §1.2, 1.3].

A6. *Self-supervised learning requires a human to label a small subset of the data.*
False. It builds its own learning signal from the observations (predict a hidden word, a masked image region); no labels are supplied [one-read §1.2].

A7. *In reinforcement learning the learner's actions change the experience it receives afterwards.*
True. Interaction means the learner partly shapes its future data, unlike the fixed datasets of supervised learning [one-read §1.2].

A8. *Whether a problem is regression or classification is decided by the type of the inputs.*
False. It is decided by the *output*: numerical output is regression, categorical output is classification, whatever the input looks like [one-read §1.3].

A9. *A prediction, a decision and an action are three different things.*
True. The prediction is what the model says about the unknown; the decision selects a response using the prediction and a criterion; the action is what is executed [one-read §1.3].

A10. *Representation learning removes the need for human choices in building a system.*
False. Humans still choose the architecture, the learning method, the experience and the objective; only more of the representation is acquired from data [one-read §1.4].

A11. *An autoregressive language model is trained by supervised learning on human-written answers.*
False. Predicting the next token from the preceding text is self-supervised: the text supplies

both context and target [one-read §1.4].

A12. *Of the four ingredients of a learning system, the evaluation criterion decides which changes to the model count as improvements.*
True. Experience, model, learning algorithm, evaluation criterion; the criterion judges success [one-read §1.5].

A13. *A system that classifies all of its training examples correctly has demonstrated generalisation.*
False. Generalisation is performance in *new* situations; training performance is only partial evidence. The wolf-on-snow system was perfect on its examples and failed in deployment [one-read §1.5].

A14. *Cross-entropy measures how good predicted probabilities are.*
True. It is the information-theoretic link: low cross-entropy means the predicted distribution puts high probability on what actually happens [one-read §1.5].

Open questions

B1. State the two ways of giving a machine an ability, and give one task where each is the right choice.

Encode human knowledge as explicit rules or procedures (right for tasks whose procedure is known and precise: computing a tax, validating a packet checksum). **Let the machine learn** from experience (right for abilities people have but cannot articulate: recognising handwriting, spotting a fraudulent transaction). The deciding question is whether useful experience exists and whether the procedure can be specified [one-read §1.1].

B2. Give the three forms of experience, the learning setting each leads to, and what provides the learning signal in each.

Examples (input with desired output) → supervised learning; signal: the labels. **Observations** with no supplied answer → unsupervised or self-supervised learning; signal: regularities in the data, or a target the method hides from itself. **Interactions** → reinforcement learning; signal: rewards that follow actions, often delayed [one-read §1.2].

B3. Labels can come from three kinds of sources. Name them and give one danger of each.

Measurement (a price, a temperature): instrument error and units. **Human or expert annotation**: disagreement, fatigue, and systematic bias of the annotators. **The future** (did the customer churn, did the patient recover): the label arrives late, and whether the label is observed may itself depend on what was done (an alerted flow gets investigated, an ignored one does not) [one-read §1.3].

B4. Describe the supervised training cycle in four steps.

(1) The current model takes a training input and produces a prediction. (2) The prediction is compared with the label, giving an error. (3) The error guides an update of the parameters. (4) The next example arrives and the cycle repeats. Training is the computational process by which experience changes the model [one-read §1.3].

B5. A hospital model outputs a probability of disease. Identify the prediction, a decision, and an action, and explain why the probability alone does not determine the action.

Prediction: the probability 0.3. Decision: whether to order a confirmatory test, which depends on the probability *and* a criterion (costs of a missed disease versus an unnecessary test). Action: the

test is ordered, or the patient is sent home. The same probability leads to different actions under different costs; the model supplies belief, not policy [one-read §1.3, 2.2].

B6. What is the difference between feature engineering and representation learning? What did deep learning change?

In feature engineering a designer chooses the features (area, location, rooms) and the model learns only how they relate to the output. In representation learning the machine discovers a useful representation from experience. Deep learning suits this because successive layers build higher-level features from lower-level ones; what changed is that more of the representation is acquired from data, not that humans disappeared [one-read §1.4].

B7. Explain the "wolves and snow" failure using the vocabulary of the chapter, and say what experiment reveals it.

The training experience contained a spurious regularity: wolves photographed on snow, dogs on grass. The system learned background instead of animal shape, scored perfectly on its examples, and failed at deployment; it did not generalise. A wolf on grass and a dog on snow reveal which cue was used. Dataset composition is therefore part of the method [one-read §1.5, 3.1].

B8. Explain in one paragraph how generation emerges from a model that was only ever trained to predict.

An autoregressive model is trained to predict the next token given the previous ones. Once trained, generation is repeated prediction: predict a token, append it to the context, predict again. Nothing beyond next-token prediction was ever learned; long coherent output is the consequence of that prediction being good across many steps [one-read §1.4].

Answer key, Chapter 2: From representations to decisions

True or false

A1. *A categorical feature with five values should be encoded as the integers 1 to 5 before being used with Euclidean distance.*
False. Integers impose an arbitrary order and spacing; a one-hot encoding puts every pair of categories at the same distance [one-read §2.1].

A2. *If the conditional distribution $P(Y \mid X = x)$ were known, the best decision would still depend on the loss.*
True. The Bayes rule minimises the conditional expected loss; squared loss gives the mean, absolute loss the median, 0–1 loss the most probable class [one-read §2.2].

A3. *Under squared loss the optimal prediction is the conditional median.*
False. It is the conditional *mean*; the median is optimal under absolute loss [one-read §2.2].

A4. *With a cost c_u per unit of under-prediction and c_o per unit of over-prediction, the optimal prediction is the quantile of $Y \mid x$ at level $c_u/(c_u + c_o)$.*
True. The asymmetric (pinball) loss is minimised at that quantile; $c_u = c_o$ recovers the median [one-read §2.2].

A5. *For a binary classifier with costs C_{FP} and C_{FN}, the optimal rule declares positive when $p(1 \mid x) > C_{FN}/(C_{FP} + C_{FN})$.*

False. The threshold is $C_{FP}/(C_{FP} + C_{FN})$: a costly miss must *lower* the threshold [one-read §2.2].

A6. *Adding a reject (review) option can never change the decision on a case that was already clearly positive.*
True. Review is optimal only in a middle band; cases with $p \geq 1 - C_R/C_{FP}$ are still declared positive [one-read §2.2].

A7. *The i.i.d. assumption says that each example is drawn from the same distribution as the deployment cases, independently of the others.*
True. It is what makes the training set a sample of the population and the empirical average an estimate of the expectation [one-read §2.3].

A8. *In k-NN, a larger k gives the model more capacity.*
False. Smaller k means more capacity: the fitted function can change at every point. $k = n$ is a constant [one-read §2.4, 3.2].

A9. *Scaling the features changes the predictions of k-NN but not those of a decision based on the true posterior.*
True. Distance, hence neighbours, depend on units; the posterior $P(Y \mid x)$ is a property of the problem and invariant to how x is written [one-read §2.4].

A10. *In high dimension, almost all of a query's n training points are at nearly the same distance from it, so "nearest" loses meaning.*
True. The radius needed to capture a few neighbours, $r_d \approx (3/n)^{1/d}$, tends to 1: neighbourhoods are no longer local [one-read §2.4].

A11. *A hypothesis space is the set of functions a parameterised model can represent.*
True. $\mathcal{H} = \{f_\theta : \theta \in \Theta\}$; the parameters index it [one-read §2.5].

A12. *The number of neighbours k is a parameter of the k-NN model.*
False. It is a *hyperparameter*: fixed before fitting and chosen on validation data, not learned from the training set [one-read §2.5].

A13. *The softmax output of a model with C classes lies in the probability simplex Δ^{C-1}.*
True. Its entries are positive and sum to one, so it is a valid distribution over the classes [one-read §2.5].

A14. *Empirical risk minimisation chooses the parameters that minimise the population risk.*
False. It minimises the *empirical* (training) risk; the population risk is unknown. The gap between the two is the subject of Lecture 3 [one-read §2.6].

A15. *Cross-entropy is used as a surrogate because the 0–1 loss has zero gradient almost everywhere.*
True. The 0–1 loss is piecewise constant in the parameters; cross-entropy is smooth, and minimising it drives the predicted probabilities toward the labels [one-read §2.6].

A16. *The validation set is used to fit the parameters and the test set to choose the hyperparameters.*
False. Training fits parameters, validation chooses hyperparameters and models, the test set is touched once for the final estimate [one-read §2.7].

Open questions

B1. Derive the Bayes decision rule for the 0–1 loss from the conditional risk.

For action a, the conditional risk is $\mathbb{E}[\mathbf{1}(Y \neq a) \mid x] = \sum_{c \neq a} P(c \mid x) = 1 - P(a \mid x)$. Minimising it over a means maximising $P(a \mid x)$: predict the most probable class. Its conditional risk is $1 - \max_c P(c \mid x)$, and the expectation over x is the Bayes error [one-read §2.2].

B2. With $C_{FP} = 1$ and $C_{FN} = 4$, derive the posterior threshold and explain why it is below 0.5.

Risk of declaring positive: $C_{FP}(1 - p) = 1 - p$. Risk of declaring negative: $C_{FN}\, p = 4p$. Positive when $1 - p < 4p$, i.e. $p > 1/5 = 0.2$. The threshold is below 0.5 because a miss costs four times a false alarm, so it is worth alerting on cases that are more likely negative than positive [one-read §2.2].

B3. Explain why two training examples with identical x and different y put a floor on the error of every classifier, and name that floor.

Any rule based on x gives the same answer to both, so one is misclassified whenever such x occur. In the population this means $0 < P(1 \mid x) < 1$ at that x and a conditional risk of $\min\{p, 1 - p\} > 0$. The floor is the Bayes error; more data or capacity cannot go below it, only a better representation can [one-read §2.2].

B4. Describe how k-NN estimates $P(Y = c \mid x)$ and what the estimate's bias and variance depend on.

$\hat{P}_k(c \mid x)$ is the fraction of the k nearest training points with label c. Small k: the neighbourhood is tight so the estimate is nearly unbiased but averages few labels, high variance. Large k: more labels averaged, low variance, but the neighbourhood stretches into regions where the posterior differs, bias. The right k depends on n and on how fast the posterior varies [one-read §2.4].

B5. Explain the curse of dimensionality in terms of a hypercube with cells of side 0.1, and what it implies for local methods.

In d dimensions a grid of side-0.1 cells has 10^d cells, so any practical sample leaves almost all cells empty, and the radius needed to find even a few neighbours grows towards the full range of the data. "Nearest" neighbours are then far away and no longer representative of x. Local methods need either low dimension, far more data, or a representation in which the data lie near a low-dimensional structure [one-read §2.4].

B6. Contrast the two strategies of the chapter, local estimation and parameterised models, in terms of what each assumes and what each needs.

Local estimation (KNN) assumes the posterior varies smoothly so that nearby points are informative; it needs a meaningful distance and enough data near every query, and it stores all the data. Parameterised models assume the relationship lies in a chosen family $\mathcal{H}$ (linear, polynomial, a network); they compress the data into parameters fitted by minimising a loss, need the family to be adequate, and generalise by the structure they impose rather than by proximity [one-read §2.4, 2.5].

B7. Write the binary cross-entropy loss for a predicted probability $\hat{p}$ and label y, and explain what minimising its average over a training set does.

$\ell = -y \ln \hat{p} - (1 - y) \ln(1 - \hat{p})$. Averaged over the training set it is the negative conditional log-likelihood of the labels under the model, so minimising it is maximum likelihood: the model is pushed to give high probability to each observed label, and confident wrong predictions are penalised heavily [one-read §2.6, 4.2].

B8. Why does a parameterised model's prediction at an input far from all training points still exist, and why is it suspect?

The fitted function $f_{\hat{\theta}}$ is defined everywhere by its formula, so it returns a value for any x. But nothing in the training objective constrained its behaviour there: many functions agree on the sample and differ elsewhere (perfect interpolation does not determine behaviour away from the data). The prediction reflects the family's inductive bias, not evidence [one-read §2.5, 3.1].

B9. Give the role of each of the three subsets and state the one rule that must never be broken.

Training set: fit the parameters. Validation set: compare models, hyperparameters, feature sets and thresholds, and choose. Test set: one final estimate of the chosen model's performance. The rule: no choice may be made using the test set, otherwise its estimate is no longer unbiased [one-read §2.7].

B10. What is a hyperparameter, how does it differ from a parameter, and give three examples from this chapter.

A hyperparameter is a setting fixed before training that shapes the hypothesis space or the fitting procedure, chosen on validation data; a parameter is learned from the training set by minimising the loss. Examples: k in KNN, the distance or the feature weights w_j, the polynomial degree, the regularisation strength [one-read §2.5].

Answer key, Chapter 3: From fitting to generalisation

True or false

A1. *The generalisation gap is the difference between the population risk and the empirical risk of the same fitted model.*
True. $G = R(\hat{f}) - \hat{R}_{\mathcal{D}}(\hat{f})$; only the empirical risk is observable [one-read §3.1].

A2. *On average, the training risk of the ERM solution is below even the best population risk achievable in the family.*
True. $\mathbb{E}[\hat{R}(\hat{\theta})] \leq R(\theta^\star) \leq \mathbb{E}[R(\hat{\theta})]$: the training number flatters twice [one-read §3.1].

A3. *A test set gives an unbiased estimate of a model's risk provided the model was fitted without it, even if the same test set was then used to choose between ten such models.*
False. The estimate is unbiased for a *fixed* model; once the test set is used to choose, the winner's score is optimistic (winner's curse) [one-read §3.1, 3.5].

A4. *For nested families $\mathcal{H}_1 \subseteq \mathcal{H}_2$, the minimum training error over $\mathcal{H}_2$ cannot exceed that over $\mathcal{H}_1$.*
True. A minimum over a superset cannot be larger; this says nothing about population risk [one-read §3.2].

A5. *Capacity is the number of parameters of the model.*
False. Capacity is how varied the functions in $\mathcal{H}$ can be; structure, constraints, regularisation, optimisation and the data all enter. KNN has no parameters yet its capacity varies with k [one-read §3.2].

A6. *Underfitting is recognised by low training error and high validation error.*
False. That pattern is *overfitting*. Underfitting shows high training *and* validation error, close together [one-read §3.2].

A7. *In the bias–variance decomposition, the noise term can be reduced by collecting more data.*
False. Noise is the irreducible variance of Y given x, a property of the problem. More data reduces variance, not noise [one-read §3.3].

A8. *Variance measures how much the fitted prediction at a fixed x changes when the training set is redrawn.*
True. $\mathrm{Var}_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(x)]$; bias is the systematic deviation of the average fit from the truth [one-read §3.3].

A9. *Approximation error is reduced by enlarging the hypothesis space; estimation error is reduced by more data.*
True. A larger family can get closer to the truth (less approximation error) but is harder to pin down from a finite sample (more estimation error) [one-read §3.3].

A10. *An L_2 penalty on the parameters reduces capacity by removing parameters from the model.*
False. No parameter is removed; the penalty restricts the parameters to a smaller nested set $\{\theta : \Omega(\theta) \leq c\}$, shrinking all coefficients smoothly [one-read §3.4].

A11. *An L_1 penalty tends to set some coefficients exactly to zero.*
True. The L_1 ball has corners on the axes, which the loss contours typically touch first; this is embedded feature selection [one-read §3.4].

A12. *Early stopping controls capacity even though the number of parameters never changes.*
True. Early iterations learn recurring patterns, late ones fit noise; keeping the best-validation checkpoint is implicit regularisation [one-read §3.4].

A13. *Data augmentation is a way of adding information about which variations should not change the output.*
True. Augmented copies encode an invariance (rotation, noise, time shift) the model should respect; it is an inductive bias supplied through the data [one-read §3.4].

A14. *Double descent means the validation error always decreases if capacity is increased far enough.*
False. It describes a second descent *beyond* the interpolation threshold in some settings; it is not a guarantee, and the first peak is real [one-read §3.3].

A15. *Forward selection and backward elimination are guaranteed to find the best feature subset.*
False. They are greedy approximations to an infeasible search over 2^d subsets; they can miss subsets whose features help only jointly [one-read §3.5].

A16. *Selecting the configuration with the lowest cross-validation error and reporting that error as the final performance is sound.*
False. The reported minimum is optimistic (winner's curse); the chosen configuration must be re-measured on an untouched test set [one-read §3.5].

A17. *A stratified split protects against near-duplicate flows from one attack episode appearing on both sides.*
False. Stratification preserves class proportions; only a *grouped* split keeps each episode entirely on one side [one-read §3.6].

A18. *A temporal split is appropriate when the deployment cases come from the future.*
True. Train on earlier data, validate and test on later data, so that drift is measured rather than hidden [one-read §3.6].

A19. *A classifier with accuracy* 0.99 *on a dataset with* 1% *positives must have caught most positives.*
False. Predicting everything negative also scores 0.99 with recall 0. Accuracy hides minority failure [one-read §3.6].

A20. *Recall and false-positive rate do not depend on the class prevalence, but precision does.*
True. Recall and FPR are computed within one class each; precision mixes the classes and falls as positives become rarer [one-read §3.6].

A21. *Computing the standardisation statistics on the full dataset before splitting is leakage even though no labels are used.*
True. Test rows influence the preprocessing; statistics must be computed on the training part only and reused unchanged [one-read §3.6].

Open questions

B1. Explain why training data are "used twice" and why this makes the training risk optimistic.

The sample $\mathcal{D}$ selects the model ($\mathcal{D} \to \hat{f}_{\mathcal{D}}$) and then evaluates it ($\hat{R}_{\mathcal{D}}(\hat{f}_{\mathcal{D}})$). Minimisation can exploit accidental regularities of that particular sample, which another sample would not share. Formally $\mathbb{E}[\hat{R}_{\mathcal{D}}(\hat{\theta}_{\mathcal{D}})] \leq R(\theta^\star) \leq \mathbb{E}[R(\hat{\theta}_{\mathcal{D}})]$, so the expected gap is non-negative and grows with capacity [one-read §3.1].

B2. List the four "doors" through which inductive bias enters a learning system, and explain why some inductive bias is unavoidable.

(1) The data and representation decide which cues exist. (2) The hypothesis space decides which relationships can be expressed. (3) Regularisation favours some fits over others. (4) The optimisation algorithm may prefer particular solutions. It is unavoidable because a finite sample is compatible with infinitely many functions that agree on the sample and differ elsewhere; without a preference, nothing singles out one of them [one-read §3.1].

B3. Write the bias–variance–noise decomposition of the expected squared error at a fixed x and define each term.

$\mathbb{E}_{\mathcal{D},Y}[(Y - \hat{f}_{\mathcal{D}}(x))^2] = \underbrace{(\bar{f}(x) - f^\star(x))^2}_{\text{bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(x) - \bar{f}(x))^2]}_{\text{variance}} + \underbrace{\text{Var}[Y \mid x]}_{\text{noise}}$, where $f^\star(x) = \mathbb{E}[Y \mid x]$ and $\bar{f}(x) = \mathbb{E}_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(x)]$. Bias is the systematic error of the average fit, variance the spread of fits across samples, noise the irreducible part [one-read §3.3].

B4. A model shows training error 0.02 and validation error 0.25. Diagnose, list three remedies, and say what you would expect from collecting ten times more data.

High variance (overfitting): the model fits the sample far better than the population. Remedies: regularise, reduce capacity, early stopping, feature selection, more representative data. With ten times more data the variance shrinks, the validation error falls and the gap closes; if capacity is adequate the training error will rise slightly as the model can no longer memorise [one-read §3.2, 3.6].

B5. Explain geometrically why L_1 regularisation produces sparse solutions and L_2 does not.

Regularised ERM is equivalent to minimising the training risk under a budget $\Omega(\theta) \leq c$. The L_2

ball is round, so the contours of the risk generically touch it at a point with all coordinates nonzero: every coefficient shrinks, none vanish. The L_1 ball is a diamond with corners on the axes; contours typically meet it first at a corner, where some coordinates are exactly zero [one-read §3.4].

B6. What is patience in early stopping, and what goes wrong if it is too short or too long?

Patience is the number of epochs training continues without validation improvement before stopping, keeping the best checkpoint. Too short: the run stops on a noise fluctuation or before a later improvement. Too long: wasted computation, though the retained checkpoint is still the best seen [one-read §3.4].

B7. Two configurations have the same population error R and validation errors $R + \varepsilon_i$ with independent $\varepsilon_i \sim N(0, \sigma^2)$. Show that the expected validation error of the selected one is $R - \sigma/\sqrt{\pi}$ and interpret.

$\min(a, b) = \frac{1}{2}(a+b-|a-b|)$ and $\varepsilon_1 - \varepsilon_2 \sim N(0, 2\sigma^2)$ with $\mathbb{E}|\varepsilon_1 - \varepsilon_2| = 2\sigma/\sqrt{\pi}$, so $\mathbb{E}[\min] = R - \sigma/\sqrt{\pi}$. Neither model is better, yet the reported number is optimistic by $\sigma/\sqrt{\pi}$; with more configurations the minimum of more noisy estimates is more optimistic still. Hence the winner must be re-measured on an untouched test set [one-read §3.5].

B8. For each of the following deployment goals, name the right split: (i) new flows from the same process; (ii) next month's traffic; (iii) new attack episodes; (iv) a rare class whose counts must be stable across folds.

(i) Random. (ii) Temporal. (iii) Grouped by episode (and by source where relevant). (iv) Stratified, combined with whichever of the others the novelty requires; stratification alone cannot override dependence between rows [one-read §3.6].

B9. Define precision, recall, specificity and F_1, and give one question each answers.

Recall $= TP/(TP+FN)$: of the real attacks, how many did we catch? Precision $= TP/(TP+FP)$: of our alarms, how many were real? Specificity $= TN/(TN+FP)$: of the benign flows, how many were left alone? $F_1 = 2PR/(P+R)$: are both precision and recall high at once? (It ignores the true negatives and weighs both error types equally.) [one-read §3.6]

B10. List five forms of leakage and give the rule that prevents the preprocessing kind.

Near-duplicates across subsets; the same device, session or patient on both sides; predictors created after the outcome or after an investigation (target leakage); preprocessing fitted on validation or test data; hyperparameters tuned on test results. Rule: compute every data-dependent transformation (scaling, imputation, selection, resampling) on the training part only and apply it unchanged to the rest, per fold in cross-validation [one-read §3.6].

B11. Describe the learning-curve patterns that indicate (a) underfitting, (b) high variance, (c) a data-limited problem, and the response to each.

(a) Both errors high and close: better representation or more capacity. (b) Training low, validation high: regularise, reduce capacity, more representative data. (c) Errors close and both still falling with n: more examples will help. A fourth pattern, test degrading over time, signals distribution shift and calls for fresh data and time-aware evaluation [one-read §3.6].

Answer key, Chapter 4: Discriminative and generative classification

True or false

A1. *Once the posterior $p(y \mid x)$ is known, the decision is easy; the hard part of classification is estimating the posterior.*
True. The Bayes rule is a lookup given the posterior and the costs; the two routes of the chapter are two ways of estimating the posterior [one-read §4.1].

A2. *A discriminative model estimates $p(x \mid y)$ and $p(y)$ and combines them with Bayes' rule.*
False. That is the generative route; a discriminative model estimates $p(y \mid x)$ or the boundary directly [one-read §4.2, 4.3].

A3. *In logistic regression the score is $w^\top x + b$ and the posterior estimate is the sigmoid of the score.*
True. $\hat{p}(1 \mid x) = \sigma(w^\top x + b)$; the score is the log-odds [one-read §4.2].

A4. *Minimising cross-entropy in logistic regression is the same as maximising the conditional likelihood of the labels.*
True. Cross-entropy is the negative conditional log-likelihood under i.i.d. examples [one-read §4.2].

A5. *Changing the cost threshold rotates the decision boundary of a logistic model.*
False. It moves the boundary parallel to itself: $w^\top x + b = \text{logit}(t)$ has the same normal vector for every t [one-read §4.2].

A6. *A generative model can in principle generate new inputs; a discriminative model cannot.*
True. The generative model has $p(x \mid y)$ to sample from; the discriminative model only has $p(y \mid x)$ [one-read §4.3, 4.6].

A7. *Maximum likelihood for a Gaussian class model sets the class mean to the sample mean and the covariance to the sample covariance of that class.*
True. Fitting $p(x \mid y = c)$ by ML gives the empirical mean and covariance of the class's examples, with the prior as the class frequency [one-read §4.3, 4.4].

A8. *LDA assumes each class has its own covariance matrix.*
False. LDA shares one covariance across classes; QDA gives each class its own [one-read §4.4].

A9. *The LDA decision boundary is linear because the quadratic terms in x cancel between classes.*
True. With a shared Σ, $x^\top \Sigma^{-1} x$ appears identically in every class score and drops out of the comparison [one-read §4.4].

A10. *In d dimensions, QDA with two classes estimates about $d(d+1)$ covariance parameters, twice as many as LDA.*
True. Each symmetric $d \times d$ covariance has $d(d+1)/2$ parameters; QDA needs one per class [one-read §4.4].

A11. *Covariance shrinkage blends the estimated covariance with a simpler matrix to reduce estimation variance.*
True. With few examples per class the full covariance is unstable or singular; shrinking towards a diagonal or scaled identity trades bias for stability [one-read §4.4].

A12. *Naive Bayes assumes that the features are independent.*
False. It assumes they are independent *given the class*; features can be strongly correlated overall through the class [one-read §4.5].

A13. *Without smoothing, one feature value never seen in a class makes that class's posterior exactly zero.*
True. A zero factor in the product vetoes all other evidence; additive (Laplace) smoothing and an "other" category prevent it [one-read §4.5].

A14. *Correlated features make Naive Bayes under-confident.*
False. They repeat the same evidence, which multiplies the odds more than once: *over*-confident posteriors [one-read §4.5].

A15. *A generative model can handle a missing feature by marginalising it out; a discriminative model needs a separate treatment.*
True. With an explicit $p(x \mid y)$ the missing coordinate can be integrated out (for Naive Bayes, drop its factor) [one-read §4.6].

A16. *A flexible discriminative model can eventually reach lower error than a generative one because it spends no capacity on aspects of x irrelevant to y.*
True. Generative structure helps with little data when its assumptions roughly hold; a wrong family leaves bias that more data cannot remove [one-read §4.6].

A17. *If the training and deployment distributions are identical, an imbalanced prior is an error that must be corrected by resampling.*
False. Imbalance without shift is not an error: the posterior should reflect it. The real problem is few minority examples, where weighting or resampling may help *estimation* [one-read §4.7].

A18. *A classifier is calibrated if, among cases given probability u, about a fraction u are positive.*
True. That is the definition; calibration is needed before cost thresholds and prior-shift corrections mean anything [one-read §4.7].

A19. *Under class-prior shift the class-conditional distributions change and the priors stay the same.*
False. The reverse: $p(x \mid y)$ is unchanged and the priors π_c change [one-read §4.7].

A20. *A high maximum posterior $\max_c \hat{p}(c \mid x)$ shows that x resembles the training data.*
False. It only says one known class outranks the others; a discriminative model can be confident far from all data. A generative likelihood check is needed for familiarity [one-read §4.8].

A21. *A per-flow classifier cannot detect an attack whose individual flows all look ordinary, but a classifier on window features can.*
True. Window features (counts, distinct ports, failure fraction, score sums over recent flows from one source) put the pattern into the representation [one-read §4.8].

A22. *An autoregressive language model is trained generatively, and a sequence-classification head on top of it is trained discriminatively.*
True. $p_\theta(w_{1:T}) = \prod_t p_\theta(w_t \mid w_{<t})$ is a generative objective; the head models $p_\psi(y \mid w_{1:T})$ [one-read §4.8].

Open questions

B1. Write the two routes to the posterior as formulas and name what each must estimate.

Discriminative: estimate $p_\theta(y \mid x)$ directly, e.g. $\sigma(w^\top x + b)$; estimate (w, b). Generative: estimate the prior $\pi_c = P(y = c)$ and the class-conditional $p_\theta(x \mid y = c)$, then $p(y = c \mid x) = \pi_c\, p_\theta(x \mid c) / \sum_k \pi_k\, p_\theta(x \mid k)$; estimate the priors and the parameters of each class model [one-read §4.2, 4.3].

B2. Explain why the sigmoid is the right link for a binary discriminative model, and the softmax for a multiclass one.

The sigmoid maps any real score to $(0, 1)$ and is the inverse of the log-odds, so modelling the log-odds as linear gives $\hat{p} = \sigma(s)$. For C classes, softmax maps C scores to a point of the simplex (positive, summing to one) and reduces to the sigmoid when $C = 2$; it is the natural generalisation of exponentiating log-odds and normalising [one-read §4.2].

B3. Starting from the Gaussian log-posterior, explain which terms are common to all classes in LDA and write the resulting score $s_c(x)$.

$\ln p(x \mid c) + \ln \pi_c = -\frac{1}{2}(x - \mu_c)^\top \Sigma^{-1}(x - \mu_c) - \frac{1}{2} \ln \det \Sigma - \frac{d}{2} \ln 2\pi + \ln \pi_c$. Expanding, $-\frac{1}{2} x^\top \Sigma^{-1} x$, $\ln \det \Sigma$ and the 2π term are identical across classes and drop out, leaving $s_c(x) = x^\top \Sigma^{-1} \mu_c - \frac{1}{2} \mu_c^\top \Sigma^{-1} \mu_c + \ln \pi_c$, linear in x [one-read §4.4].

B4. How does the class prior move the LDA boundary, and why does it not change the class models?

The prior enters only through $\ln \pi_c$, a constant in each score, so the boundary $s_0 = s_1$ translates parallel to itself, away from the rarer class. The fitted Gaussians (means, covariance) describe $p(x \mid y)$ and are unaffected; the prior only reweights them [one-read §4.4].

B5. State the Naive Bayes assumption as a formula, explain why it is attractive with little data, and give one way it fails for network-flow features.

$p(x \mid y = c) = \prod_j p(x_j \mid y = c)$. Attractive because each factor is a one-dimensional distribution estimated from counts, so the parameter count grows linearly in d and tiny classes still yield estimates. It fails when features are produced together by the same mechanism: a scanner that uses many ports also produces short failed connections, so port count and failure fraction are correlated within the attack class [one-read §4.5].

B6. Give the discriminative/generative comparison on five criteria: modelled quantity, input assumptions, missing features, sampling inputs, raw high-dimensional input.

Discriminative: posterior or boundary; fewer assumptions; missing features need separate treatment; cannot sample inputs; often practical on raw high-dimensional input. Generative: prior and class model; stronger assumptions; can sometimes marginalise missing features; can sample in principle; full density estimation is hard in high dimension [one-read §4.6].

B7. Explain the difference between correcting for class imbalance in training and correcting for prior shift at deployment.

Imbalance correction (weighting, resampling) addresses *estimation*: with few minority examples the posterior is estimated badly, and reweighting the loss can help, at the risk of distorting probabilities. Prior-shift correction addresses a *change in the world*: the posterior itself is different at deployment, and a calibrated training posterior is transformed by replacing the prior odds. One is about data scarcity, the other about a different population; both are distinct from the cost threshold, which changes the action, not the probability [one-read §4.7].

B8. Derive the prior-shift correction for a binary discriminative model from Bayes' rule, and compute it for $\pi_t = 0.2$, $\pi_d = 0.05$, $r_t(x) = 0.6$.

Posterior odds = prior odds × likelihood ratio: $\frac{r_t}{1-r_t} = \frac{\pi_t}{1-\pi_t} \cdot \frac{p(x|1)}{p(x|0)}$. Under prior shift the likelihood ratio is unchanged, so $\frac{r_d}{1-r_d} = \frac{\pi_d/(1-\pi_d)}{\pi_t/(1-\pi_t)} \cdot \frac{r_t}{1-r_t}$. Solving gives $r_d = \frac{(\pi_d/\pi_t)r_t}{(\pi_d/\pi_t)r_t + \frac{1-\pi_d}{1-\pi_t}(1-r_t)}$. Numbers: ratio 0.25, numerator 0.15, second term $\frac{0.95}{0.8} \times 0.4 = 0.475$, so $r_d = 0.15/0.625 = 0.24$ [one-read §4.7].

B9. Describe how a generative model can reject an unfamiliar input, and give one limitation of the method.

Pick $c^\star = \arg\max_c \hat{p}(c \mid x)$, then accept only if $\log p_{\hat{\theta}}(x \mid c^\star) \geq \tau_{c^\star}$, with the thresholds validated on data that reproduce the intended novelty (attack types held out). Limitation: a low likelihood is evidence of incompatibility, not proof; high-dimensional likelihoods behave oddly and a misspecified class model can reject valid cases [one-read §4.8].

B10. A detector computes, in a 60-second window ending at each flow completion, the number of flows, distinct destination ports, failure fraction and score sum. State what information is legitimately available at each decision time and give two ways such a system leaks the future.

Available: every flow from the same source completed at or before the current time, with its features and score. Leaks: using the episode's total length or end time; using later flows (a centred window); using a label or ticket produced after investigation; grouping windows so that overlapping windows of one episode fall on both sides of the split [one-read §4.8, 3.6].

B11. Give the three questions the chapter uses for choosing between a generative and a discriminative model.

Can a credible $p_\theta(x \mid y)$ be stated? Then its structure may save labelled data. Is the goal accurate prediction with many labels? Then a flexible discriminative model avoids needless assumptions about the input. Is the raw input complex and high-dimensional? Then learn a representation first and put a simple head on it. In every case, validation on deployment-like data decides [one-read §4.8].